Jerson Bonilla-Fernández

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https://github.com/JersonBonilla • https://jersonbonilla.github.io/portfolio/

EDUCATION

University of Costa Rica **March 2023 - Present**

B.S. in Computer Science | Academic Average: 9.46/10

Relevant Courses: Big Data Analysis, Robotics, Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, C/C++, C#, Java, R, SQL, MATLAB, RISC-V Assembly

Machine Learning & Data: PyTorch, scikit-learn, NLP, Transformers

Development & Tools: Git, GitHub, Android Studio, Jupyter, Google Colab

PROJECTS

Text Classification and Statistical Analysis | Python, scikit-learn, R

March 2026 - July 2026

- Implemented SVM and Multinomial Naive Bayes classifiers for text classification using TF-IDF representations.
- Conducted 400 experimental runs using the 20 Newsgroups text dataset to evaluate the effects of model selection and stopword removal on classification accuracy.
- Applied ANOVA and post-hoc analysis to assess the statistical significance of experimental factors.

RISC-V Processor Design and Pipeline Simulation | Logisim, RISC-V Assembly

March 2026 - July 2026

- Designed and simulated a dual-core RISC-V processor in Logisim, implementing a 5-stage instruction pipeline with control logic.
- Implemented registers between stages and mechanisms for control hazards, including forwarding, stalling, and bubbles.
- Implemented a spinlock synchronization mechanism for shared-memory considerations.

LingoPartner: Humanoid Robot Language Learning System | Python, Java, Pepper

August 2026 – Present

- Developing a language-learning system integrating a Pepper humanoid robot with an LLM.
- Implementing interactive language exercises including Anki-like vocabulary practice and role-playing scenarios.

A Comparative Analysis of AI-Generated and Human-Made Task Guidance Systems in Procedural Cooking Tasks

August 2026 – Present

- Researching how LLM-generated and human-created task guidance influence task performance, cognitive load, and affective response in procedural cooking tasks.
- Designing methods to compare different types of task guidance using objective and subjective measures through a within-subjects study.

LANGUAGES

Spanish: Native

English: C1 — Cambridge C1 Advanced certificate

Portuguese: C1 — Centro de Estudos Brasileiros (CEB)

French: A1 — Fundación UCR